Theseus Cyber Risk Assessment

As we work to 'death-proof' our system, the *Theseus* team performed a security assessment based on the Risk Management Framework (RMF) cycle detailed by NIST in publication 800-53.

First, we prepared the system for this assessment. This occurred during team meetings where we discussed the risks we saw in our system and prepared to mitigate.

Then, we determined the risk category of our system. Risk is defined as the likelihood of an event occurring multiplied by the impact it would have, categorized as low, medium, and high risk. Because we are operating an afloat system, the likelihood of an attack is high. While our system is infrequently online, we know our adversaries will likely attempt an exploit as soon as the presence of this system is detected. Exposure of our data, however, poses a high risk. Because the goal of our system is to detect anomalies, exposure of this data would reveal potential ship weaknesses that could be targeted by the enemy. This means that *Theseus* is a high-risk system.

After categorizing it, we were able to select the controls applicable to *Theseus*. Practically, being a high-risk system means our team must implement NIST 800-53 controls required for low, medium, and high-risk systems, which is the maximum number of controls required.

Currently, we are in the process of implementing these controls. Some recent accomplishments include successfully initiating automatic scans to identify vulnerabilities in accordance with control RA-5, information input validation detailed in SI-10, and network disconnection as outlined in SC-10. In addition, many controls, particularly in the AC, SC, PS, and PE categories are inherited from DoD protocols such as PKI access and background checks.

We hope this report helps the judge see that *Theseus* understands the importance of NIST 800-53 compliance and are committed to securing our system. We will continue to implement these controls to provide a secure afloat environment for the warfighter.